

Using conjoint and EVC to simulate the impact of the hypothetical launch price for a new product

Objective: I am a firm that has not yet launched a product. But, I would like to do a more serious job studying the economics of my launch prior to entering the market. Examples:

- I am a mature company that has a new product launch and I just want to get the strategy right before entry.
- I am an entrepreneur seeking VC financing. I want to use the simulations to make a convincing case to VCs that my new product (and business idea) will be cash-flow positive. A case based on actual marketing and economics will be a heck of a lot more convincing.
- I am a mature company contemplating how to innovate my products. I am using this simulation to assess a specific potential innovation. If the economics look good, I will move forward and get engineering to work on the new product. General Motors now links conjoint analysis (from marketing) to cost technology models (from engineering) to conduct simulations that not only consider demand and competition, but that also consider how different configurations of products lead to different cost structures. This is a very cool, but super-sophisticated way to integrate product design, competitive strategy, pricing and engineering.

Data: Since the product has not yet been launched in the market, there is obviously no data I can collect in the field to learn what demand looks like. So, instead, I will run a conjoint experiment to figure out what type of consumer demand I will face when I launch. Conjoint allows me to compute differentiation values in markets where I would not be able to do so “by hand.” This might be because many product attributes are very intangible (e.g. brand name or freshness of style of the product). If there are already incumbents, then I would also need to determine demand for their products to figure out what type of competition I will face.

Approach:

- 1) Run conjoint analysis that allows me to assess how much willingness-to-pay (WTP) consumers have for the characteristics describing the products already on the market as well as the product I am planning to launch.
- 2) Use my conjoint results to describe the current state of the market as it is *prior* to my launch. I will use the conjoint results to compute (i.e. predict) the EVCs of the actual incumbent products that are currently in the market. I can use these EVCs to trace out what demand looks like for the incumbent products. In fact, if I know their unit costs (or can make a good guess), I could even predict their profitability.
- 3) Use my conjoint results to describe the hypothetical state of the market *after* I launch my product. Now I will use the conjoint results not only to compute (i.e. predict) the EVCs of the actual incumbent products, I will also compute the EVC of my own product. Note, this means I

want to compute EVC for my product given the availability of all the other stuff that was already in the market place.

- a. I need to determine the reference product in each consumer segment. That is, I need to establish what each consumer segment would be buying if my product was not available. Note, different consumers segments may have different reference products! Note also that some consumer segments may not have sufficient WTP to justify buying any of the incumbent products. In this case, the reference product for the segment would be “not buy anything.”
 - b. In each segment, I then compute the reference value and differentiation value relative to the reference product determined in (a).
 - c. I can now use the EVCs in (b) to trace out the demand for my new product. This is very powerful. Even though I have not launched in practice, I can use the conjoint data to predict what my demand ought to look like in the event I do launch!
- 4) I can now simulate the impact of different candidate launch pricing strategies. If I launch at a given price, this will shift my competitors’ demand curves (i.e. it will change the EVCs of the incumbent products). To the extent that competitors’ demand curves shift in, competitors may be forced to adjust their prices and react. In some cases, my launch price could trigger an aggressive price war in the market, especially if my launch potentially draws a lot of demand away from a competitor. Of course, entry can also lead to higher prices. This is what has happened historically in response to generic entry into pharmaceuticals categories as well as in consumer packaged goods categories. Why is this? Classical economics predicts that more competition leads to lower prices?! But, classical economics looks at commodities. In practice, firms sell differentiated products. Suppose a generic enters and captures all of your lowest-WTP consumers. Your demand shifts in (bad). But, your remaining demand may be considerably more inelastic because you are now only facing your hard-core loyals. As a result, it could be optimal for you to respond by raising price!

Step (4) can be very sophisticated. In my own work (research and consulting) I use more sophisticated demand models than the EVC demand we studied in class. But the basic idea is exactly the same. When I simulate pricing and competitive reactions, I literally compute (on a computer) the Nash equilibrium in prices that would emerge under different launch scenarios. I do not think most pricing and/or management consulting firms are at the point they could do something this sophisticated. But, some of them could certainly do simulations along the lines of what I showed you in class. You would probably need to hire a good marketing or industrial organization professor to conduct more state-of-the-art pricing simulations.